MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

CLOCK, MICROPROCESSOR, DIGITAL, KEY CODE JRH

This amendment forms a part of Military Specification MIL-M-28787/288A(EC), dated 28 April 1980, and is approved for use by the Naval Electronic Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5 AND 6

* TABLE I, delete and substitute new table I as printed on pages 3, 4, 5, and 6 of this amendment.

PAGES 7 AND 8

6.4.1, delete and substitute the following paragraph:

"6.4.1 Input to output relationship. By providing +5 V to pin 1 and +12 V to pin 3, the "A" side of the module will provide all necessary clocking signals for the HRH module. A stable crystal controlled 2 MHz clock out of the Ø2 (TTL) output pin 5 can be provided for any system by providing 5 V to pin 1 and pin 3. The "B" side of the module provides a power fail/restart system when pins 22, 28, and 34 are attached to the HRH supplies, pin 24 is connected to pin 25 and pin 26 is connected to the RESIN input at pin 9. If any supply is disconnected or goes below ±1.0 V for the ±5 V supplies or +4 V for the +12 V supply, the appropriate FET turns on and begins to discharge capacitor C1 through resistor R2. This action will generate a high level at the RESET output (pin 12) within 500 µs of power fail for the -5 V and +12 V power supplies and within 2 ms of power fail for the software interrupt at pin 22. When power is reapplied, capacitor C1 will charge through resistor R1. This action will generate a low level at the RESET output (pin 12), no sooner than 30 ms but no later than 120 ms, after power has been reapplied. With pin 39 connected to pin 25, a power fail indicator is provided for systems using this module as a 2 MHz oscillator with only +5 V available. The output of the -5 V regulator is pin 38. It will provide 1.0 mA at -5 V. The output of the 12 V regulator is pin 36. It will provide 200 mA at 12 V. Both regulator outputs should only be used to drive the JRH or the HRH modules."

PAGE 13

* TABLE II, delete and substitute new table II as printed on page 7 of this amendment.

PAGE 14

* FIGURE 5, delete and substitute new figure 5 as printed on page 8 of this amendment.

PAGE 20

* FIGURE 8, delete and substitute new figure 8 as printed on page 9 of this amendment.

PAGE 23

* FIGURE 9, delete and substitute new figure 9 as printed on page 10 of this amendment.

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PAGE 26

TABLE VIII, delete and substitute new table VIII as printed on page 11 of this amendment.

PAGE 27

FIGURE 12, delete and substitute new figure 12 as printed on page 12 of this amendment.

PAGES 28, 29, AND 30

FIGURE 13, delete and substitute new figure 13 as printed on pages 13 and 14 of this amendment.

PAGE 32

* FIGURE 14, delete and substitute new figure 14 as printed on page 15 of this amendment.

NOTE: The margins of this amendment are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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Navy - EC

(Project 5963-N455)

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TABLE I. Module electrical requirements.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A1-A14	Isolation													
	Initial	II	S	10.0	∞	10.0	∞	10.0	∞	10.0	∞	10.0	∞	MΩ
	End-of-life	II	S	1.0	∞	1.0	∞	1.0	∞	1.0	∞	1.0	∞	MΩ
A15	Isolation	II	S	0.5	∞	0.5	∞	0.5	∞	0.5	∞	0.5	∞	MΩ
B	V_{CC} and V_{DD} maximum stress	III	S	← Module must pass remaining tests →										
C1-C3	Output short circuit current	IV	X	-10	-60	-10	-60	-10	-60	-10	-60	-10	-60	mA
D1-D3	Input clamp diode voltage	V	X	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	-0.4	-1.2	V
E1-E3	Static output voltage, low level	VI	X	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	0.00	0.45	V
F1,F4,F5	Static output voltage, high level	VII	X	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	2.4	4.5	V
F2,F3	Static output voltage, high level	VII	X	3.3	4.5	3.3	4.5	3.3	4.5	3.3	4.5	3.3	4.5	V
F6,F7	Static output voltage, high level	VII	X	9.0	10.8	9.0	10.8	9.0	10.8	9.0	10.8	9.0	10.8	V
G1,G3	Power fail/restart check (t_{PLH},-5 VPF and +12 VPF)	VIII	X	0	500	0	500	0	500	0	500	0	500	µs
G2,G4	Power fail/restart check (t_{PLH},-5 VPF and +12 VPF)	VIII	X	30	120	30	120	30	120	30	120	30	120	ms
G5	Power fail/restart check (t_{PLH}, SWJR)	VIII	X	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	0.0	2.0	ms
G6	Power fail/restart check (low level, +5 VPF)	VIII	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
G7	Power fail/restart check (high level, +5 VPF)	VIII	X	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0	3.0	5.0	V
H1,H2,H5	-5 V regulator voltage	IX	X	-4.85	-5.35	-4.80	-5.35	-4.85	-5.41	-4.72	-5.35	-4.85	-5.47	V
H3,H4	12 V regulator voltage	IX	X	11.4	12.6	11.1	12.6	11.4	12.92	10.67	12.6	11.4	13.29	V
H6	12 V regulator voltage	IX	X	11.4	12.6	11.1	12.6	11.4	12.92	10.67	12.6	11.4	13.29	V
J1	-5 V regulator ripple rejection	X	X	0	10	0	10	0	10	0	10	0	10	mVrms
J2	12 V regulator ripple rejection	X	X	0	24	0	24	0	24	0	24	0	24	mVrms
K	Output leakage current	XI	X	0	500	0	500	0	500	0	500	0	500	nA
L1-L3	Input current, low level	XII	X	-1	-270	-1	-270	-1	-270	-1	-270	-1	-270	µA
M1-M3	Input current, high level	XII	X	0	10	0	10	0	10	0	10	0	10	µA
N1	V_{DD} power supply current	XIII	X	0.01	12.0	0.01	12.0	0.01	12.0	0.01	12.0	0.01	12.0	mA
N2	V_{CC} power supply current	XIII	X	0.01	115	0.01	115	0.01	115	0.01	115	0.01	115	mA
P1, P2	Transition time (t_{TLH}), (ø1 and ø2), C_L = 50 pF	XIV	S	1	25	1	25	1	28	1	22	1	31	ns
P3-P7	Transition time (t_{TLH}), C_L = 50 pF	XIV	S	1	15	1	15	1	17	1	13	1	19	ns
Q1, Q2	Transition time (t_{THL}), (ø1 and ø2), C_L = 50 pF	XV	S	1	25	1	25	1	28	1	22	1	31	ns
Q3-Q7	Transition time (t_{THL}), C_L = 50 pF	XV	S	1	15	1	15	1	17	1	15	1	19	ns
R	ø1 pulse width check ($t_{ø1}$), C_L = 50 pF	XVI	S	90	115	90	115	90	115	90	115	90	115	ns

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See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
S	ø2 pulse width check ($t_{ø2}$), C_L = 50 pF	XVI	S	231	275	231	275	231	275	231	275	231	275	ns
T	ø1 to ø2 delay (t_{D1}), C_L = 50 pF	XVI	S	0	20	0	10	0	25	0	10	0	30	ns
U	ø2 to ø1 delay (t_{D2}), C_L = 50 pF	XVI	S	85	115	85	115	85	120	90	115	95	125	ns
V	ø1 to ø2 delay (t_{D3}), C_L = 50 pF	XVI	S	110	150	110	150	110	155	115	150	120	160	ns
W	ø2 to ø2 (TTL) delay (t_{DLH}), C_L = 50 pF	XVI	S	-5	+15	-5	+15	-5	+17	-3	+15	-1	+19	ns
X	ø2 to ø2 (TTL) delay (t_{DHL}), C_L = 50 pF	XVI	S	-5	+15	-5	+15	-5	+17	-3	+15	-1	+19	ns
Y	ø1 pulse period check (T_{CY}), C_L = 50 pF	XVI	S	495	505	495	505	495	505	495	505	495	505	ns
Z	ø2 to $\overline{STSTB}$ delay (t_{DSS}), C_L = 50 pF	XVI	S	304	334	304	334	304	334	304	334	304	334	ns
AA	$\overline{STSTB}$ pulse width check (t_{PW}), C_L = 50 pF	XVI	S	32	60	32	60	32	60	32	60	32	60	ns
BB	Ready to ø2 delay (t_{PHL}), (t_{DR1}), C_L = 50 pF	XVI	S	200	230	200	230	205	235	195	225	210	240	ns
CC	Ready to ø2 delay (t_{PHL}), (t_{DR2}), C_L = 50 pF	XVI	S	200	230	200	230	205	235	195	225	210	240	ns
DD	Reset to ø2 delay (t_{PHL}), (t_{DR3}), C_L = 50 pF	XVI	S	200	230	200	230	205	235	195	225	210	240	ns
EE1	Frequency check, 18 MHz	XVII	X	17.9	18.1	17.9	18.1	17.9	18.1	17.9	18.1	17.9	18.1	MHz
EE2	Frequency check, 2 MHz	XVII	S	1.98	2.02	1.98	2.02	1.98	2.02	1.98	2.02	1.98	2.02	MHz
FF1, FF2	Power supply transient amplitude (V_{CC} and V_{DD})	XVIII	Q	0	400	0	400	0	400	0	400	0	400	mV

See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test condition	Parameter	Ref test table	Test type 1/-3/	25°C ±5°C 4/		Class I 4/				Class II 4/				Unit
						0°C ±5°C		60°C ±5°C		-55°C ±5°C		85°C ±5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
GG1-GG3	Output crosstalk, low level	XIX	Q	0	500	0	500	0	500	0	500	0	500	mV
GG4, GG7	Output crosstalk, low level	XIX	Q	0	850	0	850	0	850	0	850	0	850	mV
GG5, GG6	Output crosstalk, low level	XIX	Q	0	1,000	0	1,000	0	1,000	0	1,000	0	1,000	mV
HH1-HH7	Output crosstalk, high level	XX	Q	0	400	0	400	0	400	0	400	0	400	mV
JJ1, JJ2	Power supply filter capacitance	XXI	S	0.71	1.31	0.71	1.31	0.71	1.31	0.71	1.31	0.71	1.31	µF
KK1-KK3	Input capacitance	XXII	Q	2	15	2	15	2	15	2	15	2	15	pF

1/ "X" parameters shall be 100 percent tested during module acceptance tests.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

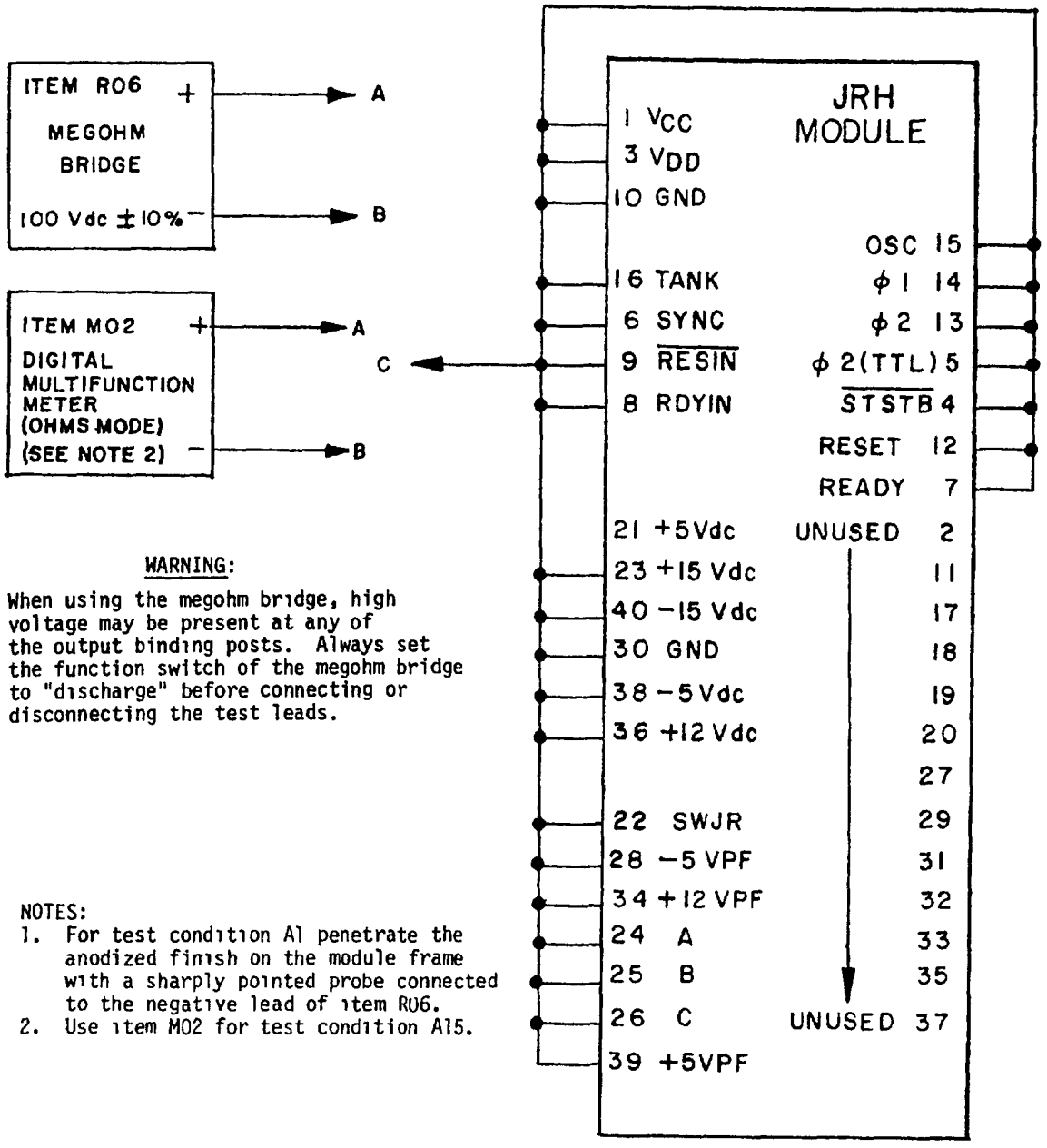
4/ Unless otherwise specified, limits are for initial and end-of-life.

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TABLE II. Isolation test - Continued.

Test condition	Parameter	Connect		Initial limits 25°C ±5°C ambient		Unit	Test figure
		A to pin	B to pin	Min	Max		
A10	Isolation	31	2,11,17,18,19, 20,27,29,32,33, 35,37	10.0	∞	MΩ	5
A11	"	32	2,11,17,18,19, 20,27,29,31,33, 35,37	"	"	"	"
A12	"	33	2,11,17,18,19, 20,27,29,31,32, 35,37	"	"	"	"
A13	"	35	2,11,17,18,19, 20,27,29,31,32, 33,37	"	"	"	"
A14	"	37	2,11,17,18,19, 20,27,29,31,32, 33,35	"	"	"	"
A15	"	1	21	0.5	"	"	"

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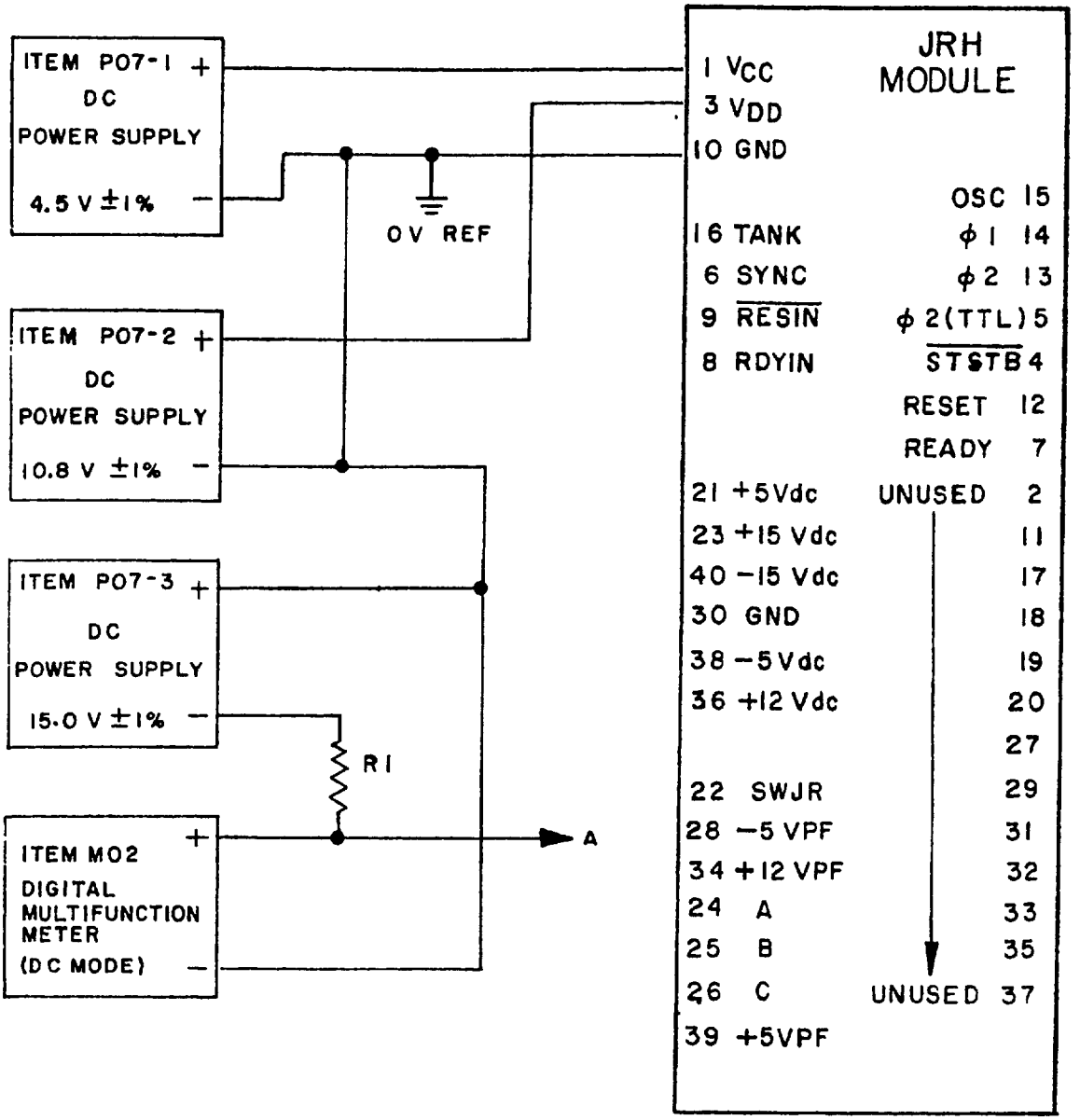
WARNING:

When using the megohm bridge, high voltage may be present at any of the output binding posts. Always set the function switch of the megohm bridge to "discharge" before connecting or disconnecting the test leads.

NOTES:
1. For test condition A1 penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative lead of item R06.
2. Use item M02 for test condition A15.

FIGURE 5. Isolation test circuit for table II.

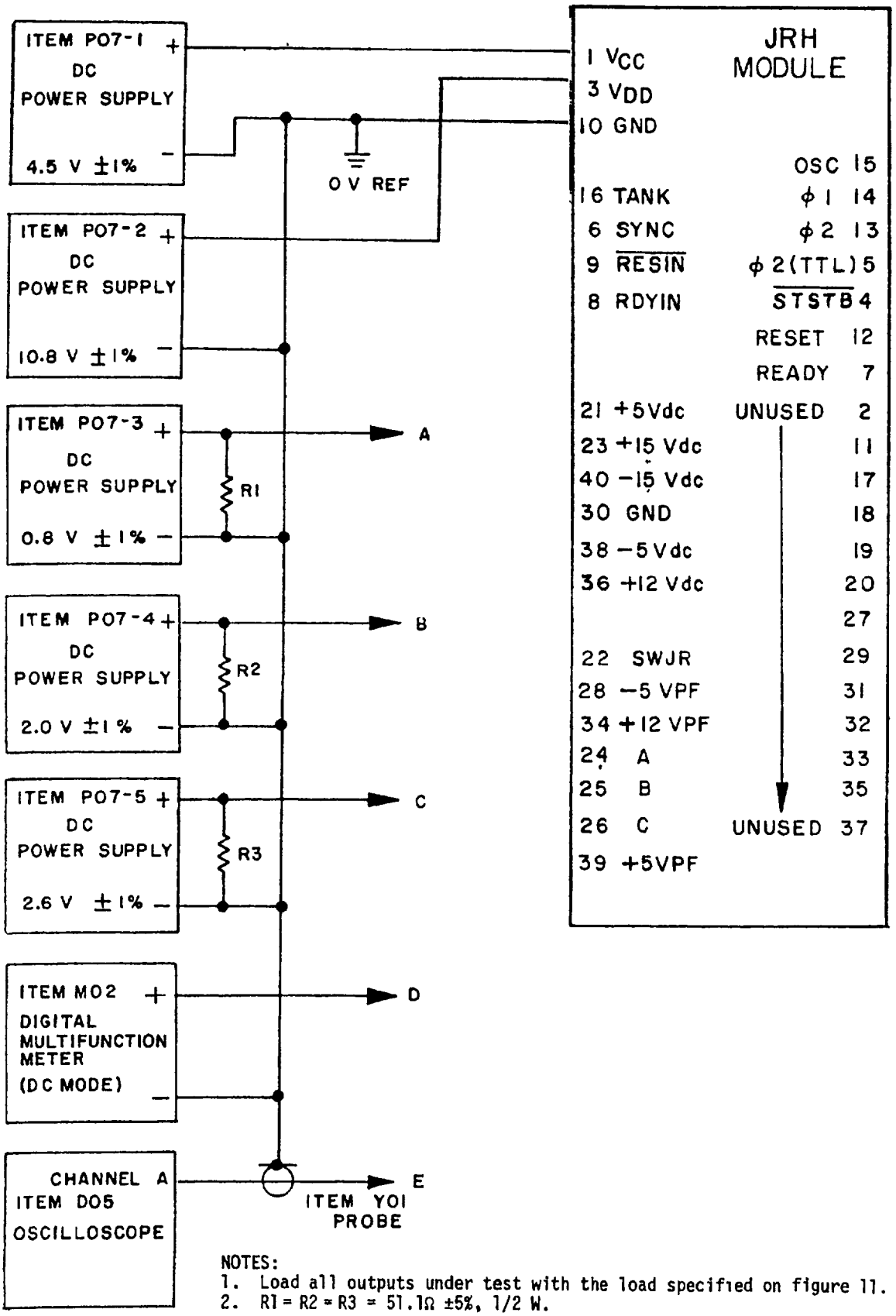
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NOTE: R = 2.74 kΩ ±1%, 1/8 W.

FIGURE 8. Input clamp diode voltage test circuit for table V.

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NOTES:
1. Load all outputs under test with the load specified on figure 11.
2. R1 = R2 = R3 = 51.1Ω ±5%, 1/2 W.

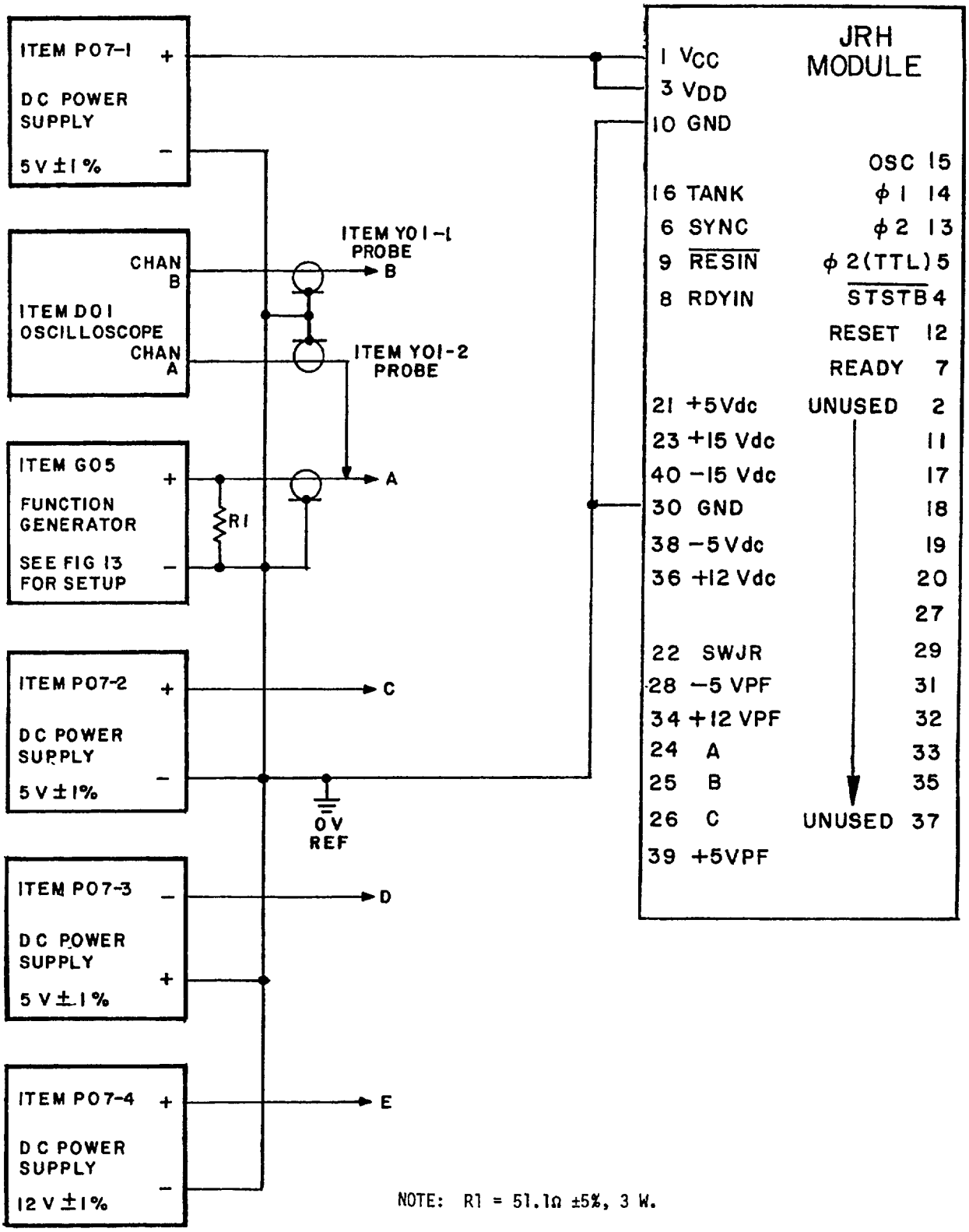
FIGURE 9. Static output voltage (low-level and high-level) test circuit for tables VI and VII.

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TABLE VIII. Power fail/restart check test.

Test condition	Parameter	Connect							Initial limits 25°C±5°C ambient		Unit	Test figure
		A to pin	B to pin	C to pin	D to pin	E to pin	Pin 25 to pin	Pin 26 to pin	Min	Max		
G1	-5 VPF Power fail/ restart check, t_{PLH}	28	12	21	Open	34	24	9	0	500	s	12,13
G2	-5 VPF Power fail, restart check, t_{PHL}	28	"	"	Open	34	"	"	30	120	ms	"
G3	+12 VPF Power fail/ restart check, t_{PLH}	34	"	"	28	Open	"	"	0	500	µs	"
G4	+12 VPF Power fail/ restart check, t_{PHL}	34	"	"	28	Open	"	"	30	120	ms	"
G5	SWJR Power fail/ restart check, t_{PLH}	22	"	"	28	34	"	"	0.0	2.0	ms	"
G6	+5 VPF Power fail/ restart check, low level	21	26	Open	Open	Open	39	B	0.0	1.5	V	"
G7	+5 VPF Power fail/ restart check, high level	21	26	Open	Open	Open	39	B	3.0	5.0	V	"

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NOTE: R1 = 51.1Ω ±5%, 3 W.

FIGURE 12. Power fail/restart check test circuit for table VIII.

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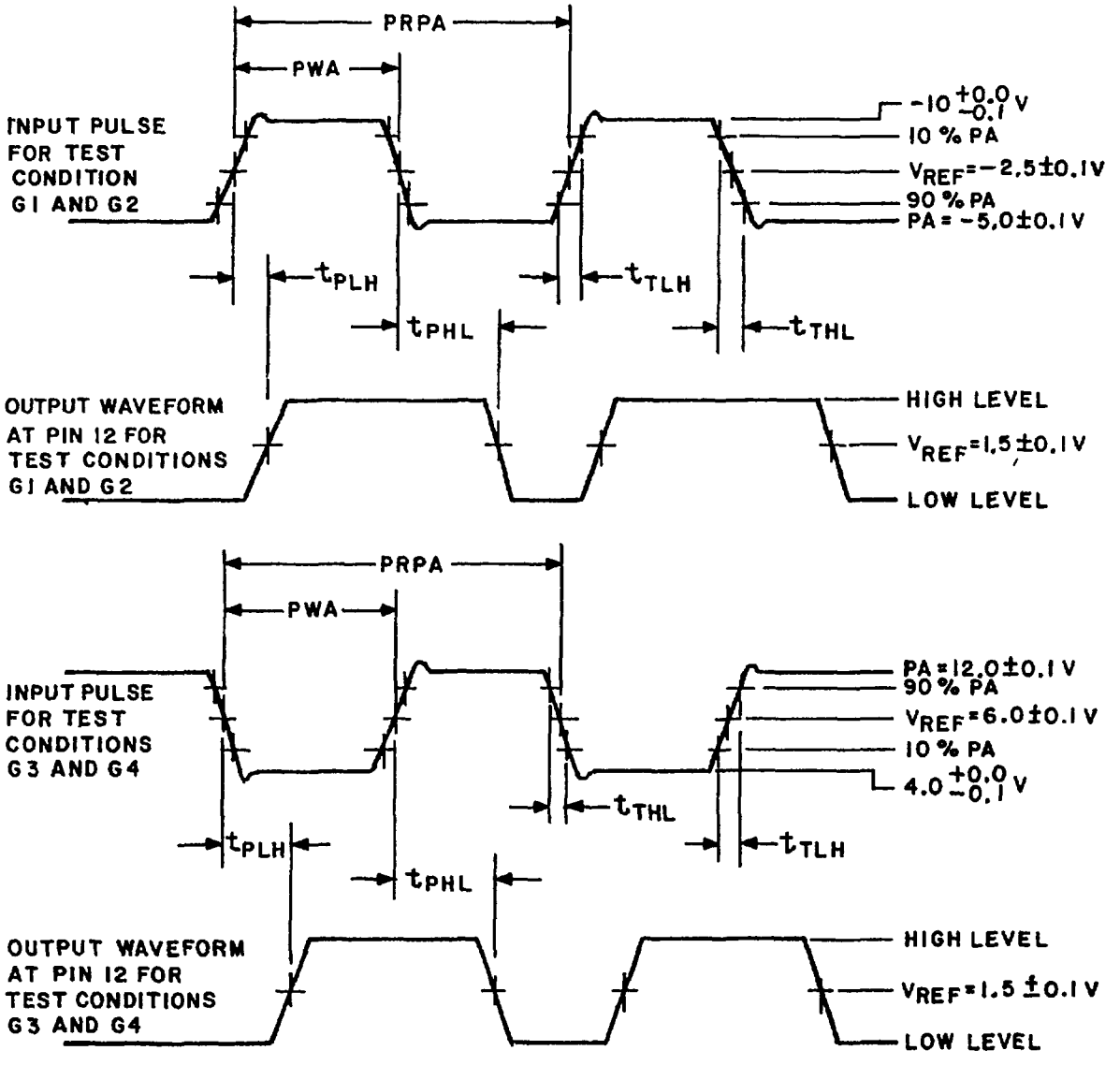
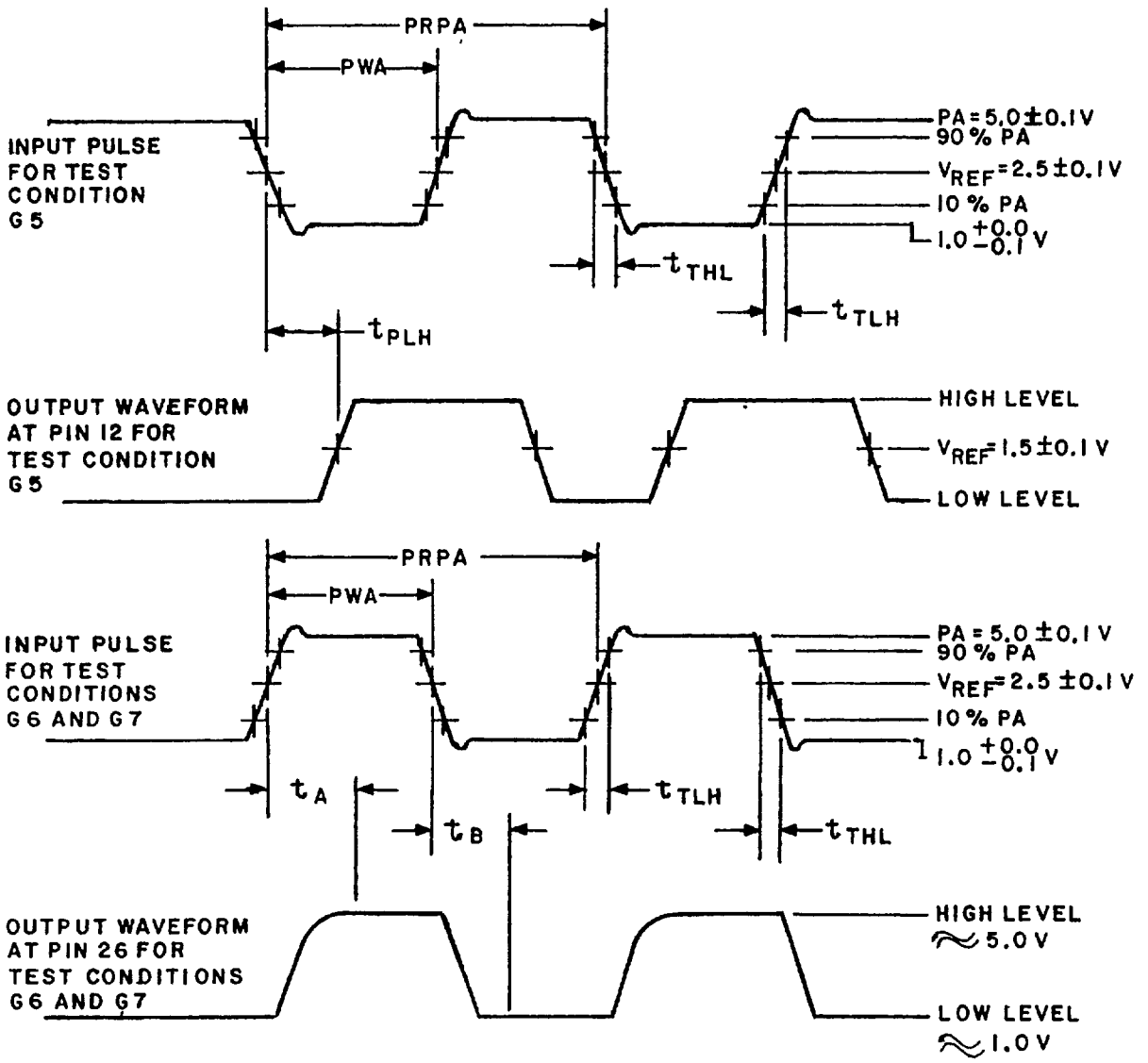


FIGURE 13. Power fail/restart check test circuit waveforms for table VIII.

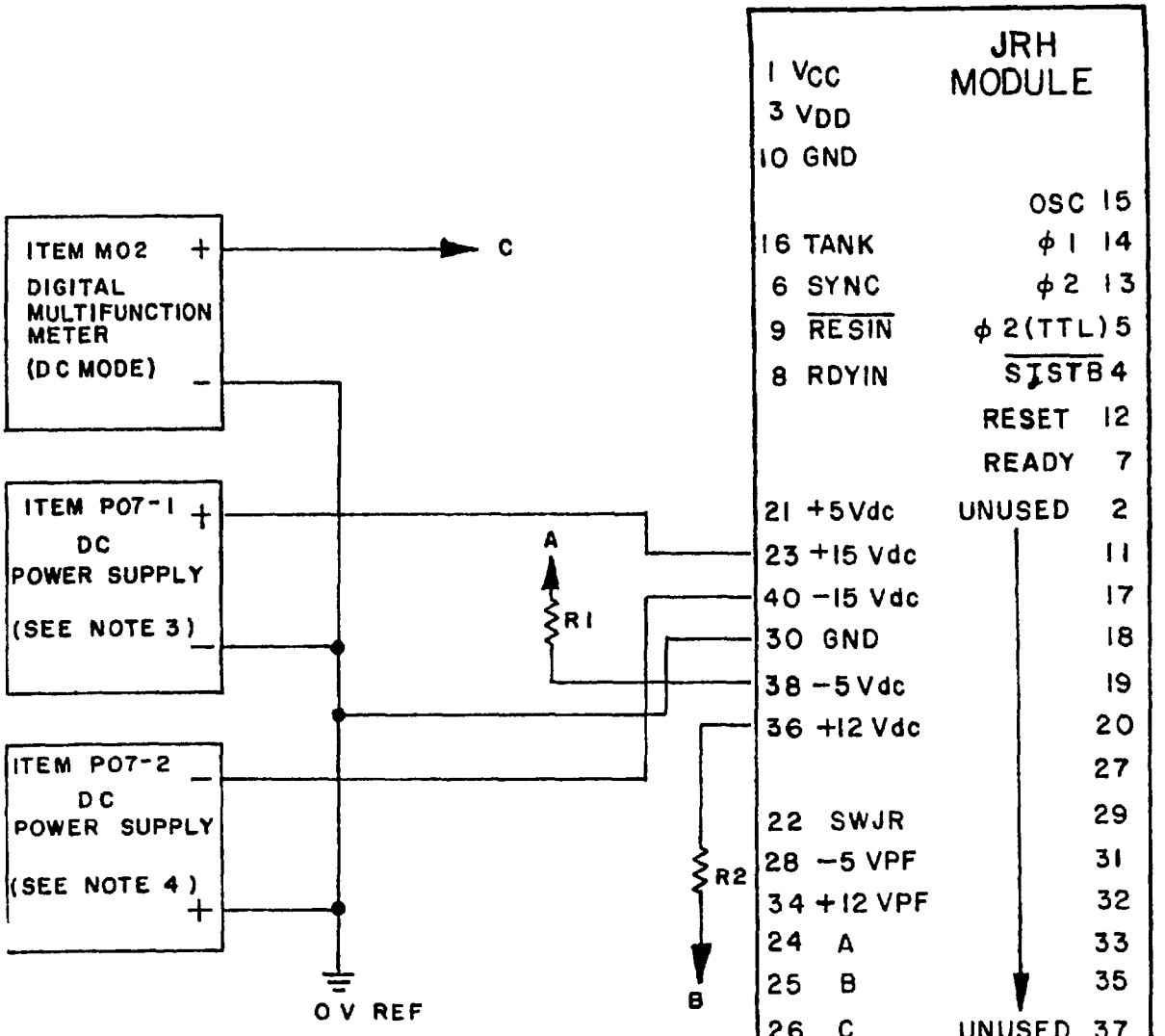
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NOTES:
1. Input pulse parameters for test conditions G1-G4:
   PRPA = 400 ±40 ms.
   PWA = 200 ±20 ms.
   $t_{TLH} = t_{THL} \leq 200$ ns.
2. Input pulse parameters for test conditions G5-G7:
   PRPA = 1,000 ±100 ms.
   PWA = 500 ±50 ms.
   $t_{TLH} = t_{THL} \leq 200$ ns.
3. For test condition G6, measure the voltage of the output waveform 3.0 ms (t_B = 3.0 ms) after the falling edge of the input pulse.
4. For test condition G7, measure the voltage of the output waveform 260 ms (t_A = 260 ms) after the rising edge of the input pulse.

FIGURE 13. Power fail/restart check test circuit waveforms for table VIII - Continued.

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NOTES:
1. R1 = 4.99 kΩ ±1%, 1/8 W.
2. R2 = 60.4Ω ±1%, 3 W.
3. For test conditions:
   H3 and H4 set item P07-1 voltage = +15.0 V ±1%.
   H6 set item P07-1 voltage = +16.5 V $^{+0}_{-1}$ %.
4. For test conditions:
   H1 and H2 set item P07-2 voltage = -15.0 V ±1%.
   H5 set item P07-2 voltage = -16.5 V $^{+0}_{-1}$ %.

FIGURE 14. Regulator output voltage (-5 V and +12 V) test circuit for table IX.

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